

# Management of pain & discomfort in dental office

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- *Dental anxiety and phobia result in avoidance of dental care. It is a frequently encountered problem in dental offices.*
  - *Formulating acceptable evidence-based therapies for such patients is essential, or else they can be a considerable source of stress for the dentist.*
  - *The initial interaction between the dentist and the patient can reveal the presence of anxiety, fear, and phobia. In such situations, subjective evaluation and self-reporting on fear and anxiety scales and objective assessment of blood pressure, pulse rate, pulse oximetry, finger temperature, and galvanic skin response can greatly enhance the diagnosis and enable categorization of these individuals as mildly, moderately, or highly anxious or dental phobics.*



Broadly, dental anxiety can be managed by psychotherapeutic interventions, pharmacological interventions, or a combination of both, depending on the level of dental anxiety, patient characteristics, and clinical situations.

- Pharmacologically, these patients can be managed using either sedation or general anesthesia.

- psychological therapies are efficient on a long-term basis with positive effects on the patients, enabling them to seek dental care in future, which should be the primary focus of the dental team.

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- *Occurrence of pain during procedures may be low (i.e., affecting less than 25% of general dental practice patients, according to Tickle et al. 2012).*
  - *However, public perceptions (deep common concepts ) are that dental care is highly associated with pain.*
  - *Consequently, dental care-related anxiety and fear are closely related with oral pain*

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- *The unpleasantness of pain, its cause(s), and future implications contribute to the emotional reaction, which is used as a cue (signal/reason) to either escape the situation and/or to avoid it in the future, if possible.*
  - *The reaction to pain differs from patient to patient and from time to time in the same patient.*

# Factors affecting an individual's perception of pain

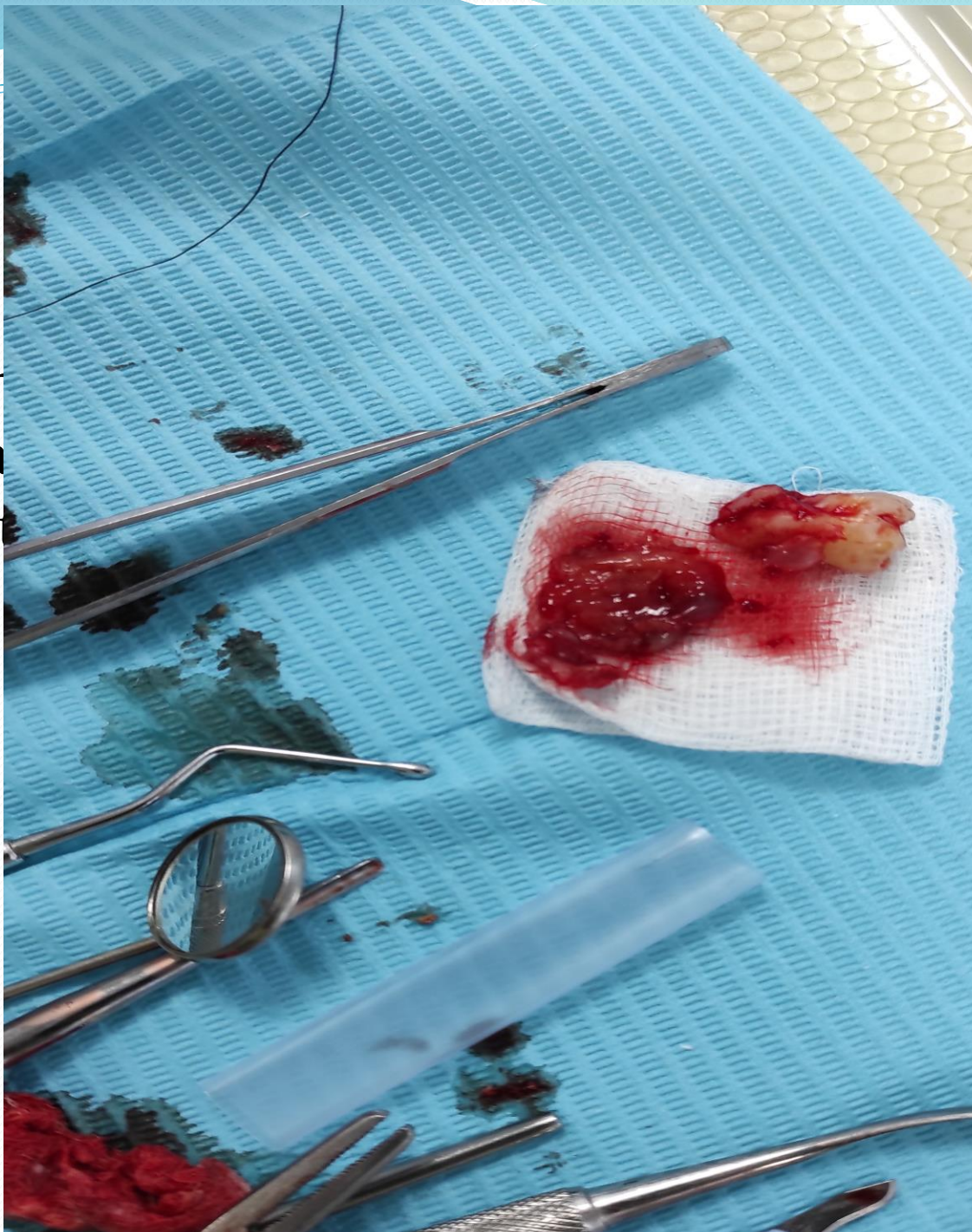
**The experience and expression of orofacial pain is determined by a multitude of factors that can be**

- Biological (e.g., innervation, genetics)
- Environmental (e.g., stimulus characteristics of the dental operatory, provider communication)
- Emotional (e.g., anxiety, fear, stress, depression)
- Cognitive (e.g., catastrophizing, memory) components.  
(catastrophizing means :to imagine the worst possible outcome of an action or event)
- Finally, patient characteristics (e.g., age, gender, race, culture, SES, and other sociocultural factors) affect the experience and expression of dental pain.



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## **Dentist-patient relationship**

**the relationship between dental patient and oral healthcare provider significantly impact the experience and expression of pain.**

- An important component of the provider–patient relationship is communication. One study demonstrated that patient attitudes about the quality of a past appointment were significantly different between patients who discussed an orofacial problem with a healthcare provider versus with a layperson (e.g., spouse, relative, friend).
- Interestingly, those patients who perceived poorer quality relationships with healthcare providers, and were less likely to discuss a pain problem with the provider, were at greater risk for developing health problems associated with not receiving appropriate dental care



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- Two facets of communication content have been found to have an effect on pain responding: provider confidence and provider behaviour that is described as “warm and friendly”
  - Patients indicated that when providers communicated confidence in an intervention, they reported less pain than when the provider was unsure or hesitated about the treatment approach.
  - Therefore, communication quality and content of communications between patient and provider, like other aspects of the overall environment in the dental office, affects not only the relationship but the experience and expression of pain as well.

# Distraction

- Distraction, or shifting the patient's attention away from painful situation to another more pleasant stimulus, has long been successful in lessening pain . Due to the acute and transient nature of procedure-related dental pain, the use of distraction often is quite successful, as it moves the focus away from the current environment.
- Distraction may be as simple as listening to music or as technologically advanced as virtual reality. it found that watching a television program or playing a video game was more effective at reducing patient discomfort than listening to an audiotape.

Both simple and complex virtual reality systems have been implemented as a means of distraction.

# Predictability

- Providing information about both the potential procedural and sensory experience appears to be the most beneficial for patients.
- While more information generally increases predictability and decreases the experience of pain and anxiety, some individuals prefer less information to cope with potentially painful situations.

# Emotional Determinants

- Anxiety, fear, stress, panic, and depression, among others, are potentially important in the experience and expression of pain.
- prevention or reduction of anxiety and fear, either pharmacologically, behaviorally, or in combination, improving mood, and focusing attention away from the pain, can and effectively decrease their pain experience.

# Cognitive Determinants

- Cognitive processes that establish, maintain, and alter memories of dental pain are critically important in terms of patients' perceptions of the dental experience and their future plans for prophylactic and other return visits.
- Dental patients with high anxiety tend to have greater expectations and recall of pain
- Moderate (conscious) sedation during oral surgery has the desired effect of being associated with memory of less pain and anxiety one month after the procedure

# Individual patient characteristics and perception of pain

- A variety of individual differences affect response to (potential and actual) pain in the dental operator.
- Several studies have concluded that patients with higher levels of dental anxiety and fear responded with more sensitivity to pain (i.e., less pain tolerance) during treatment are more likely to report dental pain, and have greater likelihood of irregular dental attendance



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- Recently researchers have found that certain genetic expressions seem to be related to greater pain sensitivity. Individuals with variants of the melanocortin-1 receptor (MC1R) gene, which are associated with red hair color, seem to have more dental anxiety and fear of dental pain and greater pain sensitivity than those without the gene variant.

# patient characteristics as a factor

- Age, gender, socioeconomic status (SES), and culture all interact with the environment, thus impacting learning history, which influences pain experience and expression.
- Younger individuals are more likely to report pain, while older adults of high SES are less likely to report pain.
- Women USUALLY report more fear of dental pain, while males reported less such fear.
- Individuals from lower SES backgrounds may have less experience with dental care from a young age and therefore may be more uncertain of what to expect during dental treatment.
- Patients of lower SES report greater orofacial pain and have greater behavioral impact from the pain as compared to those with high SES.

Thanks  
Our **future dentists** who will  
successfully treat the patients **painlessly**  
And alleviate their **fear and anxiety**

